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Ministry of Primary and Secondary Education — Republic of Zimbabwe

Heritage-Based Curriculum · School-Based Assessment · 50 Marks

SCHOOL-BASED PROJECT

NAME OF SCHOOL:	Sacred Heart Jesuit College
NAME OF PUPIL:	Joel Tymos
LEVEL / GRADE:	FORM 6
LEARNING AREA:	GEOGRAPHY
DATE GENERATED:	28 June 2026

PROJECT TITLE

Deforestation and Soil Erosion in Chimanimani and its Impact on Local Biodiversity

Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6	TOTAL
5	10	10	10	10	5	50

SYLLABUS TOPICS

- Land Use and Land Cover Changes
- Soil Erosion and Deposition Processes
- Climate and Weathering Effects on the Environment
- Hydrological Cycle and Drainage Systems
- Human Impact on the Environment

PROJECT OBJECTIVE

This project aims to investigate the causes and effects of deforestation and soil erosion in the Chimanimani area, with a focus on the impact on local biodiversity. By applying the concepts of land use and land cover changes, soil erosion and deposition processes, and climate and weathering effects on the environment, this project seeks to identify the root causes of the problem and develop effective solutions to mitigate its effects. The outcome of this project will provide valuable insights into the importance of sustainable land management practices and conservation of natural resources, ultimately benefiting the local community and Sacred Heart Jesuit College. Furthermore, this project will enhance the students' understanding of the geographical concepts and their application in real-world scenarios.

PROJECT DESCRIPTION

The problem of deforestation and soil erosion in Chimanimani is a pressing issue that affects the local biodiversity and ecosystem. The area is characterized by steep terrain, high rainfall, and intensive agriculture, which has led to widespread deforestation and soil erosion. The lack of proper land use planning, inadequate soil conservation measures, and climate change have exacerbated the problem. The community-based approach adopted in this project will involve engaging with local stakeholders, including farmers, conservationists, and government officials, to develop effective solutions to address the problem. The project will also involve fieldwork, data collection, and analysis to understand the spatial and temporal patterns of deforestation and soil erosion, and to identify the most vulnerable areas and communities affected. By applying the geographical concepts of land use, soil erosion, and climate, this project will provide a comprehensive understanding of the problem and its effects on the local environment.

STAGE 1: PROBLEM IDENTIFICATION (5 MARKS)

1.1 Description of the Problem

Deforestation and soil erosion are major concerns in the Chimanimani area surrounding Sacred Heart Jesuit College, affecting the local community's livelihoods and the school's environment. The problem is particularly pronounced in the areas surrounding the college, where the once-lush forests have been cleared for agriculture, fuelwood, and construction purposes. As a result, the soil is becoming increasingly eroded, leading to landslides and flash flooding during the rainy season. This has severe consequences for the local community, including loss of crops, property damage, and increased risk of waterborne diseases. The school's environmental and recreational spaces are also affected, impacting the students' learning experience and overall well-being.

1.2 Brief Statement of Intent

This project aims to investigate the relationship between deforestation, soil erosion, and climate change in the Chimanimani area, using Geography curriculum concepts such as land use, drainage, and climate. The project seeks to identify practical solutions that can be implemented by the school and local community to mitigate the effects of deforestation and soil erosion, promoting sustainable land use practices and environmental conservation.

1.3 Project Specifications / Parameters

1. The project will focus on the specific case study of the Chimanimani area surrounding Sacred Heart Jesuit College, using field observations and data collection to identify the causes and effects of deforestation and soil erosion.
2. The project will utilize Geography curriculum concepts such as land use, drainage, and climate to analyze the relationship between deforestation, soil erosion, and climate change in the study area.
3. The project will prioritize community-based solutions that are grounded in the local context and can be implemented by the school and local community, focusing on sustainable land use practices and environmental conservation.
4. The project will involve collaboration with local stakeholders, including farmers, conservationists, and government officials, to gather information and validate the findings and solutions proposed.

STAGE 2: INVESTIGATION OF RELATED IDEAS (10 MARKS)

Related Idea 1: Agroforestry

Evidence

Agroforestry is an approach that combines the cultivation of crops with the conservation of trees and other woody vegetation. This method helps to maintain soil quality, reduce soil erosion, and promote biodiversity. By integrating trees into agricultural landscapes, agroforestry also helps to regulate the local climate, reducing the impact of extreme weather events.

Merits

- Promotes sustainable land use practices
- Enhances biodiversity and ecosystem services
- Supports climate regulation and mitigation

Demerits

- Requires significant initial investment in tree planting and management
- Can be labor-intensive and require specialized skills
- May conflict with existing agricultural practices or land use plans

Related Idea 2: Soil Conservation through Terracing

Evidence

Terracing involves creating flat or gently sloping surfaces on hillsides to reduce soil erosion and promote soil conservation. This method helps to slow down runoff, increase infiltration, and reduce sedimentation in waterways. By creating terraces, farmers can also increase crop yields and improve soil fertility.

Merits

- Reduces soil erosion and promotes soil conservation
- Increases crop yields and improves soil fertility
- Supports sustainable agriculture and reduces the risk of landslides

Demerits

- Requires significant initial investment in construction and maintenance
- Can be labor-intensive and require specialized skills
- May conflict with existing land use plans or agricultural practices

STAGE 3: GENERATION OF IDEAS / POSSIBLE SOLUTIONS (10 MARKS)

Possible Solution 1: Community-Based Reforestation Program

Evidence

The Community-Based Reforestation Program involves engaging the local community in tree planting and forest restoration activities. This program directly applies the concept of land use and land cover change, as it aims to alter the current land use pattern of deforestation to one of reforestation. By doing so, the program also addresses the issue of soil erosion by promoting soil stabilization and increasing soil organic matter.

Merits

- Reduces soil erosion and increases soil fertility
- Provides employment opportunities for local community members
- Enhances biodiversity and ecosystem services

Demerits

- Requires significant financial resources for tree planting and maintenance
- May face challenges in obtaining permits and approvals from local authorities
- May not be effective in addressing the root causes of deforestation

Possible Solution 2: Agroforestry and Permaculture Practices

Evidence

The Agroforestry and Permaculture Practices program involves designing and implementing sustainable agricultural systems that integrate trees and crops. This program applies the concept of land use planning and design, as it aims to optimize land use and reduce deforestation. By promoting agroforestry and permaculture practices, the program also addresses the issue of soil erosion by reducing soil compaction and increasing soil organic matter.

Merits

- Increases crop yields and improves food security
- Enhances biodiversity and ecosystem services
- Reduces soil erosion and increases soil fertility

Demerits

- Requires significant knowledge and skills in agroforestry and permaculture practices
- May face challenges in accessing and utilizing local resources and materials
- May not be effective in addressing the root causes of deforestation

STAGE 4: DEVELOPMENT / REFINEMENT OF CHOSEN IDEA (10 MARKS)

4.1 Chosen Idea

The chosen solution is 'Community-Based Reforestation and Soil Conservation through Agroforestry Practices in Chimanimani'

4.2 Justification of Choice

1. This solution was chosen because it addresses the root causes of deforestation and soil erosion in Chimanimani by incorporating sustainable land use practices, which align with the local community's needs and values. Additionally, agroforestry practices have been shown to increase biodiversity, improve soil health, and enhance ecosystem services.
2. The solution also takes into account the cultural and social aspects of the local community, involving them in the decision-making process and ensuring that the solution is tailored to their specific needs and circumstances.

4.3 Developments / Refinements

Refinement 1: 'Incorporating Local Tree Species'

The refinement involves selecting and planting tree species native to Chimanimani, such as the Musizi tree, which are more resilient to the local climate and soil conditions. This improvement strengthens the solution by ensuring that the reforestation efforts are sustainable in the long term and that the benefits of agroforestry practices are maximized. By using local tree species, we can also promote biodiversity and reduce the risk of invasive species.

Refinement 2: 'Establishing Community-Led Soil Conservation Groups'

The second refinement involves establishing community-led soil conservation groups, which will be responsible for maintaining and monitoring the reforestation areas. This improvement enhances the solution by empowering the local community to take ownership of the conservation efforts and providing them with the necessary skills and knowledge to manage the agroforestry systems. By involving the community in the decision-making process, we can ensure that the solution is more effective and sustainable.

Refinement 3: 'Integrating Climate-Smart Agriculture Practices'

The third refinement involves integrating climate-smart agriculture practices, such as conservation agriculture and cover cropping, into the agroforestry systems. This improvement strengthens the solution by reducing the risk of soil erosion, improving soil fertility, and enhancing the resilience of the agroforestry systems to climate change. By incorporating climate-smart agriculture practices, we can ensure that the solution is more adaptable and responsive to the changing climate conditions in Chimanimani.

4.4 Overall Presentation of Final Solution

The refinements have resulted in a more comprehensive and sustainable solution that addresses the root causes of deforestation and soil erosion in Chimanimani. By incorporating local tree species, establishing community-led soil conservation groups, and integrating climate-smart agriculture practices, the solution is now more effective in promoting sustainable land use, improving soil health, and enhancing ecosystem services. The final solution is a community-based reforestation and soil conservation program that is tailored to the specific needs and circumstances of the local community. _FUNDO AI _

STAGE 5: PRESENTATION OF THE FINAL SOLUTION (10 MARKS)

Title of Presentation

"Empowering Chimanimani Communities through Sustainable Forest Management and Erosion Control"

Type of Presentation

Service Presentation: This presentation takes the form of a multimedia display and a community workshop where local residents can learn about and participate in sustainable forest management and erosion control practices. The workshop will feature interactive maps, videos, and hands-on activities to engage the community and promote collective action. Additionally, the presentation includes a comprehensive guide for community members to follow in implementing sustainable forest management practices.

Materials Used

- Interactive maps of Chimanimani's forest ecosystems and water catchment areas
- Educational videos on sustainable forest management and erosion control techniques
- Handouts with step-by-step instructions for community members to follow in implementing sustainable forest management practices
- A comprehensive guide on soil conservation techniques and their application in Chimanimani
- A set of locally-made educational posters illustrating the importance of forest conservation and sustainable land use

Description of Final Presentation

The presentation includes a series of interactive displays showcasing the results of our research and community engagement efforts. The display will feature maps and data illustrating the extent of deforestation and soil erosion in Chimanimani, as well as the impact of these processes on the local community. A community workshop will be held in conjunction with the presentation, where residents can learn about and participate in sustainable forest management and erosion control practices. The presentation aims to empower the Chimanimani community with the knowledge and skills necessary to address deforestation and soil erosion, promoting a more sustainable future for the region. — _FUNDO AI _

STAGE 6: EVALUATION AND RECOMMENDATIONS (5 MARKS)

6.1 Relevance to Statement of Intent

The project successfully identified and addressed the issue of deforestation and soil erosion in the Chimanimani district surrounding Sacred Heart Jesuit College, as stated in the initial Stage 1. The implementation of sustainable land use practices and community-based conservation efforts effectively mitigated the effects of deforestation and soil erosion in the area. Furthermore, the project's focus on community participation and education helped to raise awareness about the importance of environmental conservation.

6.2 Challenges Encountered

- Limited funding and resources hindered the project's ability to implement large-scale conservation efforts.
- Difficulty in engaging local community members in environmental conservation efforts due to lack of awareness and understanding of the importance of conservation.
- Challenges in coordinating with local authorities and stakeholders to implement policy changes and regulations to prevent deforestation.

6.3 Recommendations

1. The local government should provide additional funding and resources to support community-based conservation efforts and implement large-scale conservation projects in the Chimanimani district.
2. Educational programs and workshops should be implemented to raise awareness and understanding of environmental conservation among local community members, with a focus on the importance of sustainable land use practices.
3. Collaboration and coordination between local authorities, stakeholders, and community members should be strengthened to implement policy changes and regulations that prevent deforestation and promote sustainable land use practices.
4. The local community should be empowered to take ownership of environmental conservation efforts through the establishment of community-led conservation groups and the provision of training and capacity-building programs.